

Maize is a **C4** plant.

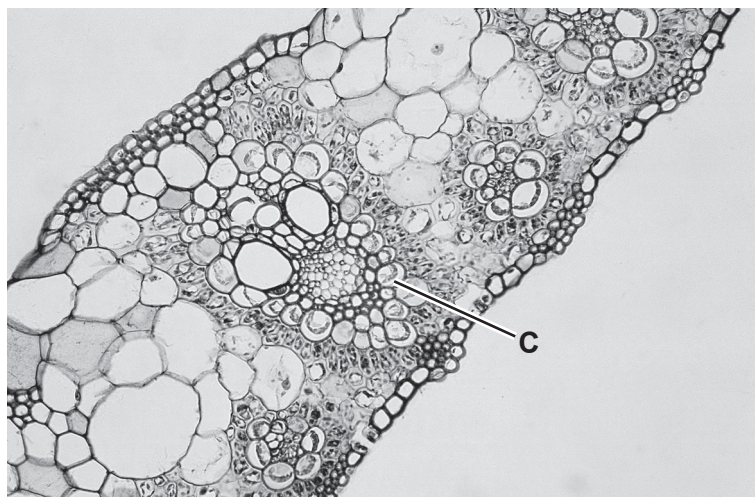


Fig. 1.1

(a) On Fig. 1.1, use label lines and letters to show:

A – a cell in the epidermal layer

B – a cell that contains PEP carboxylase.

[2]

(b) (i) Identify the cell type labelled **C** in Fig. 1.1.

C [1]

(ii) Explain how the leaf anatomy shown in Fig. 1.1 adapts the C4 plant to maintain a high rate of photosynthesis at high temperatures.

[4]

[4]

- The rate of CO₂ uptake is used to measure the rate of photosynthesis.



- [4]

- (ii) Using Fig. 1.2, compare the rates of photosynthesis in low and high CO₂ conditions in the C4 plant between 30 °C and 35 °C **and** suggest an explanation for this difference.

.....

.....

.....

.....

.....

..... [2]

[Total: 13]